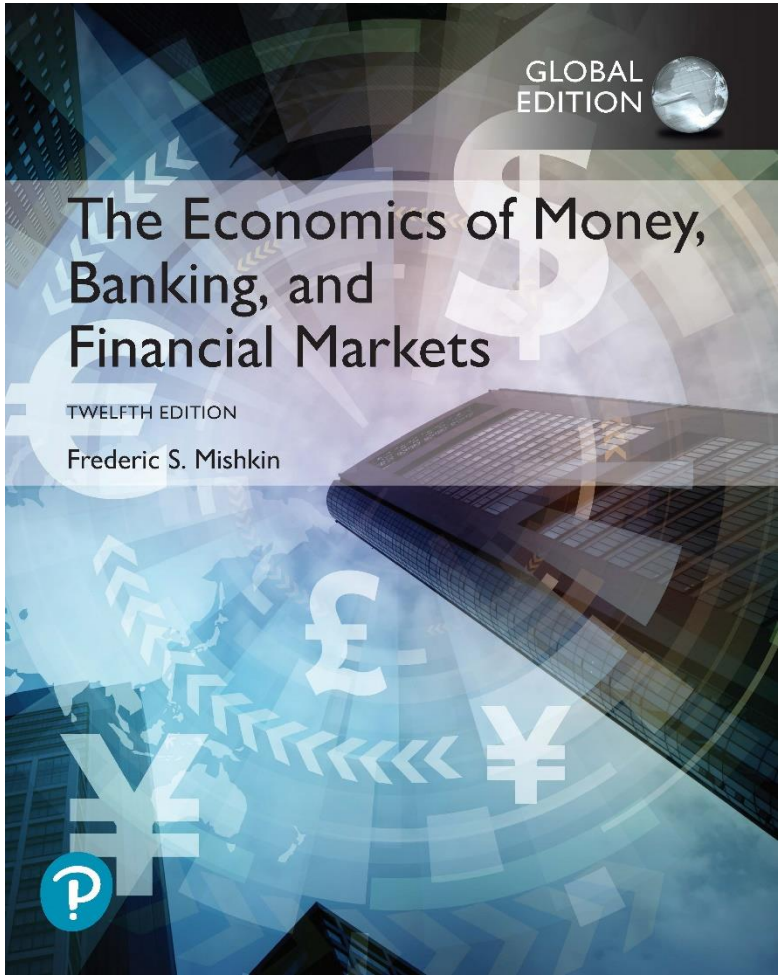
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 11

Banking Industry: Structure and Competition

Preview

- This chapter examines the historical trends in the banking industry that help explain the unique structure of the U.S. system.

Learning Objectives (1 of 2)

- Recognize the key features of the banking system and the historical context of the implementation of these features.
- Explain how financial innovation led to the growth of the shadow banking system.
- Identify the key structural changes in the commercial banking industry.
- Summarize the factors that led to consolidation in the commercial banking industry.

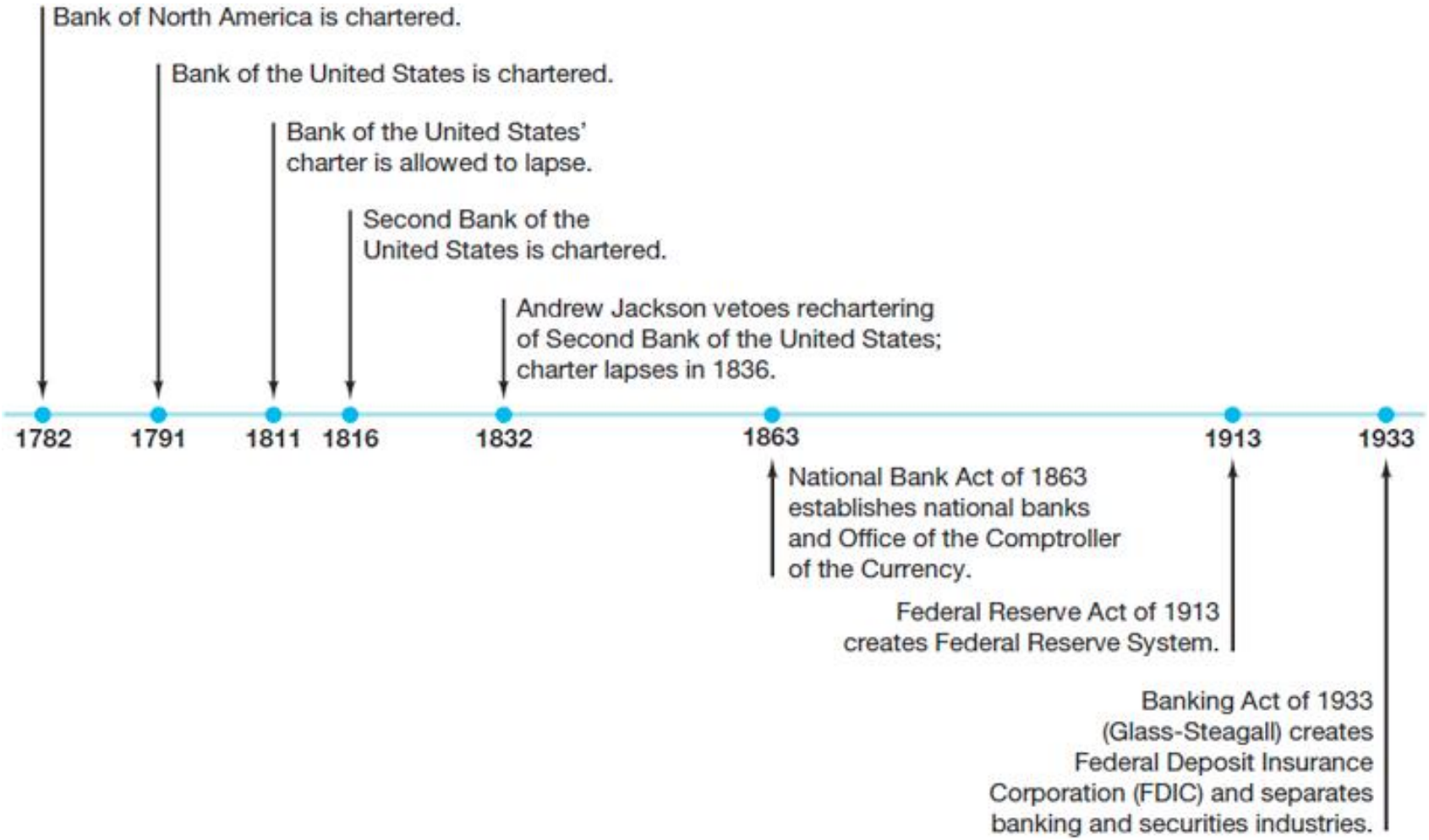
Learning Objectives (2 of 2)

- Assess the reasons for separating banking from other financial services through legislation.
- Summarize the distinctions between thrift institutions and commercial banks.
- Identify the reasons for U.S. banks to operate in foreign countries and for foreign banks to operate in the United States.

Historical Development of the Banking System

- Bank of North America chartered in 1782
- Controversy over the chartering of banks
- National Bank Act of 1863 creates a new banking system of federally chartered banks
  - Office of the Comptroller of the Currency
  - Dual banking system
- Federal Reserve System is created in 1913.

Figure 1 Time Line of the Early History of Commercial Banking in the United States



Primary Supervisory Responsibility of Bank Regulatory Agencies

- Federal Reserve and state banking authorities: state banks that are members of the Federal Reserve System.
- Fed also regulates bank holding companies.
- FDIC: insured state banks that are not Fed members.
- State banking authorities: state banks without FDIC insurance.

Financial Innovation and the Growth of the "Shadow Banking System"

- Financial innovation is driven by the desire to earn profits
- A change in the financial environment will stimulate a search by financial institutions for innovations that are likely to be profitable
  - **Financial engineering**

Responses to Changes in Demand Conditions: Interest-Rate Volatility

- **Adjustable-rate mortgages**
  - Flexible interest rates keep profits high when rates rise
  - Lower initial interest rates make them attractive to home buyers
- **Financial derivatives**
  - Ability to hedge interest rate risk
  - Payoffs are linked to previously issued (i.e., derived from) securities.

Responses to Changes in Supply Conditions: Information Technology

- **Bank credit and debit cards**
  - Improved computer technology lowers transaction costs
- **Electronic banking**
  - ATM, home banking, ABM, and virtual banking
- **Junk bonds**
- **Commercial paper market**

Securitization and the Shadow Banking System

- **Securitization**
  - To transform otherwise illiquid financial assets into marketable capital market securities.
  - Securitization played an especially prominent role in the development of the subprime mortgage market in the mid 2000s.

Bruce Bent and the Money Market Mutual Fund Panic of 2008

- Bruce Bent, one of the originators of money market mutual funds, almost brought down the industry during the global financial crisis in the fall of 2008.
- Not surprisingly, given the extension of a government safety net to the money market mutual fund industry, there are calls to regulate this industry more heavily.

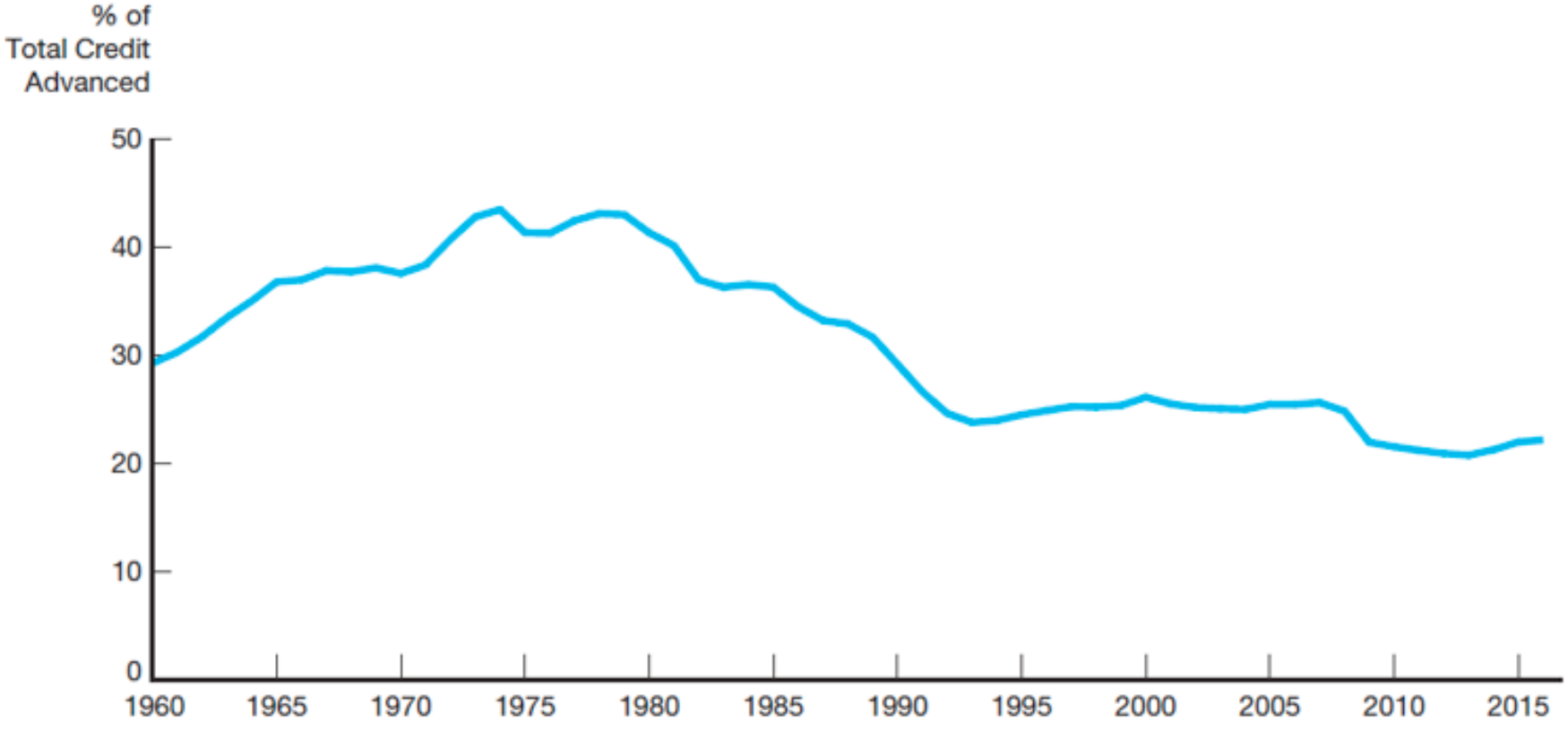
Avoidance of Existing Regulations

- Loophole Mining:
  - Reserve requirements act as a tax on deposits
  - Restrictions on interest paid on deposits led to disintermediation
  - Money market mutual funds
  - Sweep accounts

Financial Innovation and the Decline of Traditional Banking (1 of 3)

- As a source of funds for borrowers, market share has fallen.
- Commercial banks' share of total financial intermediary assets has fallen
- No decline in overall profitability
- Increase in income from off-balance-sheet activities

Figure 2 Bank Share of Total Nonfinancial Borrowing, 1960–2017



Source: Federal Reserve Bank of St. Louis, FRED data base: http://research.stlouisfed.org/fred2/; https://www2.fdic.gov/hsob/index.asp.

Financial Innovation and the Decline of Traditional Banking (2 of 3)

- Decline in cost advantages in acquiring funds (liabilities)
  - Rising inflation led to rise in interest rates and disintermediation
  - Low-cost source of funds, checkable deposits, declined in importance
- Decline in income advantages on uses of funds (assets)
  - Information technology has decreased need for banks to finance short-term credit needs or to issue loans
  - Information technology has lowered transaction costs for other financial institutions, increasing competition

Financial Innovation and the Decline of Traditional Banking (3 of 3)

- Banks' Responses
  - Expand into new and riskier areas of lending
    - Commercial real estate loans
    - Corporate takeovers and leveraged buyouts
  - Pursue off-balance-sheet activities
    - Non interest income
    - Concerns about risk
    - Decline of Traditional Banking in Other Industrialized Countries
- Decline of Traditional Banking in Other Industrialized Countries

Structure of the U.S. Commercial Banking Industry

- Restrictions on branching
  - McFadden Act and state branching regulations
- Response to branching restrictions
  - Bank holding companies
  - Automated teller machines

Table 1 Size Distribution of Insured Commercial Banks, March 31, 2017

Table 1 Size Distribution of FDIC Insured Banks, March 31, 2017

Assets	Number of Banks	Share of Banks (%)	Share of Assets Held (%)
Less than $100 million	1,501	25.6	0.5
$100 million–$1 billion	3,605	61.6	6.9
$1 billion–$10 billion	632	10.8	10.4
$10 billion–$250 billion	109	1.9	31.6
More than $250 billion	9	0.2	50.6
Total	5,856	100.00	100.00

Source: FDIC Quarterly Banking Profile, https://www.fdic.gov/bank/analytical/qbp/index.html

Table 2 Ten Largest U.S. Banks, 2017

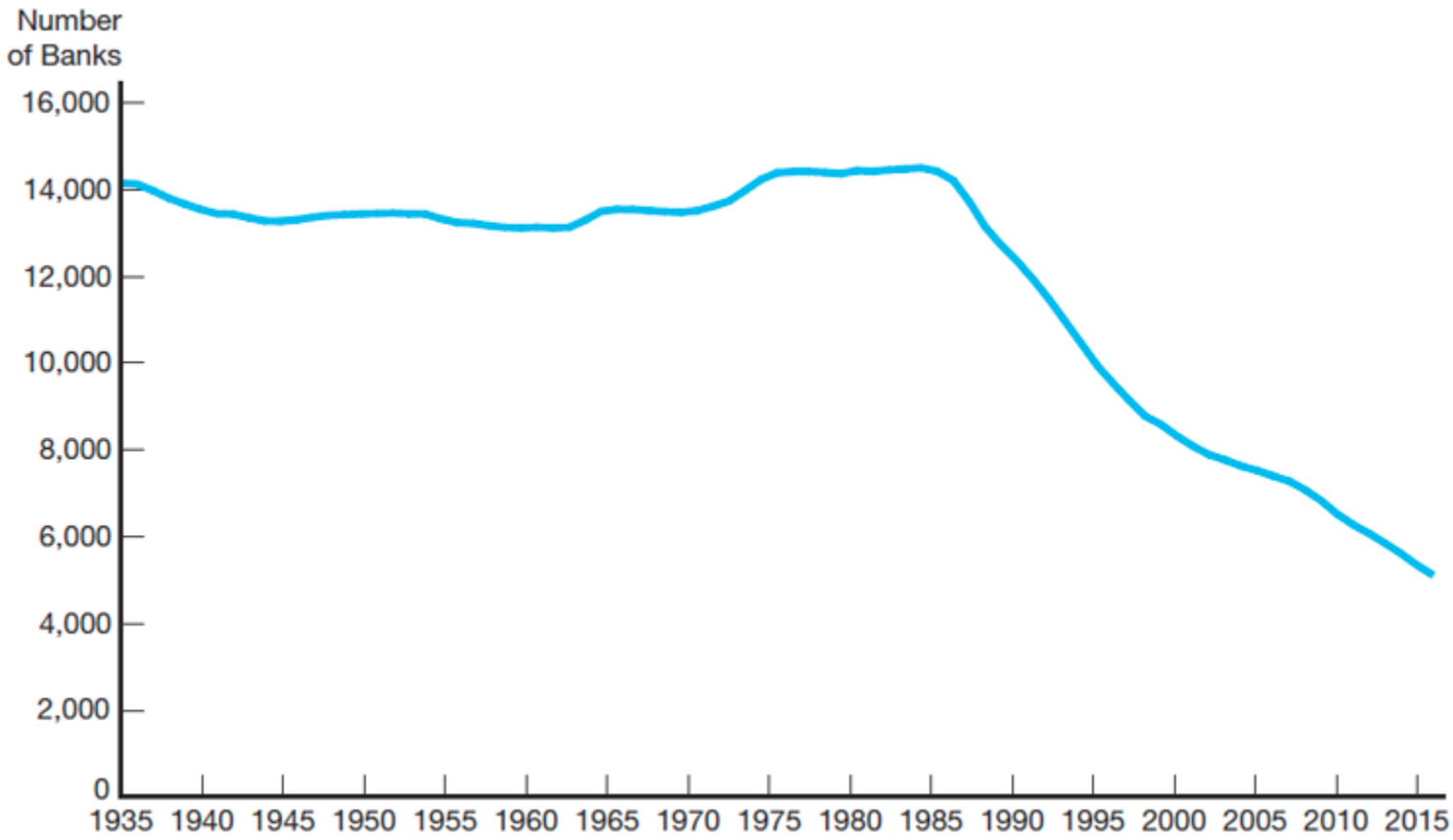
Bank	Assets ($ billions)	Share of All Commercial Bank Assets (%)
1. J.P. Morgan Chase & Co.	2,420	14.4
2. Bank of America Corp.	2,150	12.8
3. Citigroup Inc.	1,770	10.5
4. Wells Fargo & Co.	1,750	10.4
5. U.S. Bankcorp	416	2.5
6. Bank of New York Mellon Corp.	377	2.2
7. PNC Financial Services Group	362	2.2
8. Capital One Financial Corp.	314	1.9
9. HSBC North America Holdings	292	1.7
10. TD Bank U.S. Holding Co.	253	1.5
Total	10,103	60.1

Source: From Bankrate.com—Compare mortgage, refinance, insurance, CD rates: http://www.bankrate.com/banking/americas-top-10-biggest-banks/#slide=1.

Bank Consolidation and Nationwide Banking

- The number of banks has declined dramatically over the last 30 years.
  - Bank failures and consolidation
  - Deregulation: Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994
  - Economies of scale and scope from information technology
- Results may be not only a smaller number of banks but a shift in assets to much larger banks.

Figure 3 Number of Insured Commercial Banks in the United States, 1934–2016 (Third Quarter)



Source: Federal Reserve Bank of St. Louis, FRED database: http://research.stlouisfed.org/fred2/.

What Will the Structure of the U.S. Banking Industry Look Like in the Future?

- Although the United States retains a unique banking structure in possessing a large number of banks, its structure is converging with systems in Europe and Japan.
- How far the convergence between banking systems will extend is the subject of ongoing academic debate

Are Bank Consolidation and Nationwide Banking Good Things? (1 of 2)

- **Benefits**
  - Increased competition, driving inefficient banks out of business
  - Also, increased efficiency from economies of scale and scope
  - Lower probability of bank failure from more diversified portfolios

Are Bank Consolidation and Nationwide Banking Good Things? (2 of 2)

- **Costs**
  - Elimination of community banks may lead to less lending to small business
  - Banks expanding into new areas may take increased risks and fail

Large bank (versus small banks)

Large bank (versus small banks)

-- cheap funding due to scale effect

-- hierarchy, hard to deal with soft information

centralized decision needs hard information

Stein (JF, 2002)

This paper asks how well different organizational structures perform in terms of generating information about investment projects and allocating capital to these projects. A decentralized approach—with small, single-manager firms—is most likely to be attractive when information about projects is "soft" and cannot be credibly transmitted. In contrast, large hierarchies perform better when information can be costlessly "hardened" and passed along inside the firm. The model can be used to think about the consequences of consolidation in the banking industry, particularly the documented tendency for mergers to lead to declines in small-business lending.

Separation of the Banking and Other Financial Service Industries (1 of 2)

- **Erosion of Glass-Steagall**
  - Prohibited commercial banks from underwriting corporate securities or engaging in brokerage activities
  - Section 20 loophole was allowed by the Federal Reserve enabling affiliates of approved commercial banks to underwrite securities as long as the revenue did not exceed a specified amount
  - U.S. Supreme Court validated the Fed's action in 1988

Separation of the Banking and Other Financial Service Industries (2 of 2)

- **Gramm-Leach-Bliley Financial Services Modernization Act of 1999**
  - Abolishes Glass-Steagall
  - States regulate insurance activities
  - SEC keeps oversight of securities activities
  - Office of the Comptroller of the Currency regulates bank subsidiaries engaged in securities underwriting
  - Federal Reserve oversees bank holding companies
  - Stimulated consolidation of the banking industry

Separation of Banking and Other Financial Services Industries Throughout the World

- No separation between banking and securities industries in Germany, Holland and Switzerland
- Universal banking in Great Britain
  - May engage in security underwriting
  - Separate legal subsidiaries are common
  - Bank equity holdings of commercial firms are less common
  - Few combinations of banking and insurance firms
- Japanese style system

The Global Financial Crisis and the Demise of Large, Free-Standing Investment Banks

- Although the move toward bringing financial service activities into larger, more complex banking organizations was inevitable after the demise of Glass-Steagall, no one expected it to occur as rapidly as it did in 2008. Over a six-month period from March to September 2008, all five of the largest, free-standing investment banks ceased to exist in their old form.

Thrift Industry: Regulation and Structure

(1 of 3)

- **Savings and loan associations**
  - Chartered by the federal government or by states
  - Most are members of Federal Home Loan Bank System (FHLBS)
  - Deposit insurance provided by Savings Association Insurance Fund (SAIF), part of FDIC
  - Regulated by the Office of Thrift Supervision

Thrift Industry: Regulation and Structure
(2 of 3)

- **Mutual savings banks**
  - Approximately half are chartered by states
  - Regulated by state in which they are located
  - Deposit insurance provided by FDIC or state insurance

Thrift Industry: Regulation and Structure
(3 of 3)

- **Credit unions**
  - Tax-exempt
  - Chartered by federal government or by states
  - Regulated by the National Credit Union Administration (NCUA)
  - Deposit insurance provided by National Credit Union Share Insurance Fund (NCUSIF)

International Banking

- **Rapid growth**
  - Growth in international trade and multinational corporations
  - Global investment banking is very profitable
  - Ability to tap into the Eurodollar market

Eurodollar Market

- Dollar-denominated deposits held in banks outside of the United States
- Ironic birth in Communist period
- Most widely used currency in international trade
- Offshore deposits not subject to regulations
- Important source of funds for U.S. banks

Structure of U.S. Banking Overseas

- **Shell operation**
- **Edge Act corporation**
- **International banking facilities** (**IBFs**)
  - Not subject to regulation and taxes
  - May not make loans to domestic residents

Foreign Banks in the United States (1 of 2)

- Agency office of the foreign bank
  - Can lend and transfer fund in the United States
  - Cannot accept deposits from domestic residents
  - Not subject to regulations
- Subsidiary U.S. bank
  - Subject to U.S. regulations
  - Owned by a foreign bank

Foreign Banks in the United States (2 of 2)

- Branch of a foreign bank
  - May open branches only in state designated as home state or in state that allow entry of out-of-state banks
  - Limited service may be allowed in any other state
- Subject to the International Banking Act of 1978

Table 3 Ten Largest Banks in the World, 2017

Bank	Assets (U.S. $ trillions)
1. Industrial and Commercial Bank of China, China	3.62
2. China Construction Bank Corp., China	2.94
3. Agricultural Bank of China, China	2.82
4. Bank of China, China	2.63
5. HSBC Holdings, United Kingdom	2.57
6. JP Morgan Chase & Co., US	2.42
7. BNP Paribas, France	2.40
8. Mitsubishi UFJ Financial Group, Japan	2.32
9. Bank of America, US	2.15
10. Credit Agricole Group, France	1.91

Source: From Bankrate.com—Compare mortgage, refinance, insurance, CD rates, http://www.bankrate.com/finance/banking/largest-banks-in-the-world-1.aspx.